On Activation Plateaus

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Executive Summary

Recent work has shown that spherical interpolation (slerp) between certain prompt pairs exhibits "activation plateaus": regions where model output stays constant despite activation changes, separated by sharp boundaries. The natural interpretation is that these prompts lie in different stable regions of activation space. We test this by asking: if genuinely distinct regions exist, can any smooth path connect them?

We define an optimization objective balancing path length against output volatility (analogous to least action) and search for smooth paths between prompt pairs that show sharp boundaries along slerp. For "The house was big" → "The house was in", we find a nearly linear output profile (Figure 1, blue), but the path is approximately 20 times longer than slerp. We also analyze the

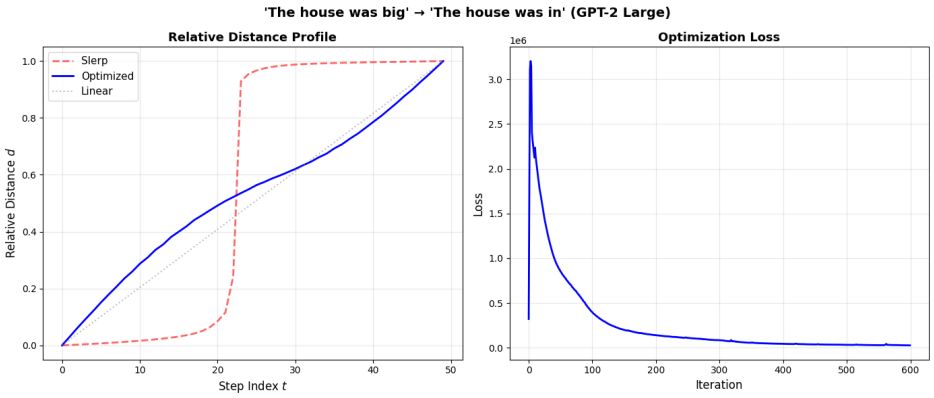


Figure 1: Relative distance along slerp (red) versus optimized path (blue) for "The house was big" → "The house was in". Slerp shows sharp plateau-boundary-plateau pattern; the optimized path is nearly linear.

eigenspectrum of the Jacobian at plateau and boundary points along the slerp path. At all three points tested, the spectrum is sparse: one singular value is ∼10× larger than the second. The top singular value at the boundary is ∼10× larger than at the plateaus. The interpolation direction aligns with the top singular vector at the boundary (cosine similarity 0.66) but not at the start (0.06).

1 Introduction

This document summarizes an investigation following up on the LessWrong post "Activation Plateaus: Where and How They Emerge" by Matthew Shinkle & Stefan Heimersheim. They

showed showed that spherical interpolation (slerp) between certain prompt pairs exhibits activation plateaus: regions where model output remains nearly constant despite changes in activation, separated by sharp boundaries where output changes rapidly.

The interpretation offered is that prompts' activations lie in different "stable regions" of activation space. But if genuinely distinct stable regions exist, no smooth path should connect them. Any path would have to cross a boundary. Slerp shows a boundary, but is it possible to find a path that doesn't?

2 Path Optimization Framework

2.1 Motivation

Slerp is a natural choice for interpolation, but it is not the only possible path between two points in activation space. If the "stable regions" interpretation is correct, then the boundary observed along slerp should be unavoidable: every path between the two prompts should cross it. Finding a smooth path would suggest that the stable-regions picture needs refinement.

2.2 The Optimization Objective

We define the optimal path $P(t)$ between points A and B in activation space as the curve minimizing the following functional:

$$J(P) = \int_0^1 \|P'(t)\|^2 \, dt + \lambda \int_0^1 \left\| \frac{d}{dt} f(P(t)) \right\|^2 dt \tag{1}$$

subject to boundary conditions $P(0) = A$ and $P(1) = B$.

The first term penalizes path length in activation space ("input kinetic energy"). The second term penalizes output volatility, i.e. how much the model output f changes along the path ("output kinetic energy"). The parameter λ balances these objectives:

- At $\lambda = 0$, we recover a straight line (shortest path, ignoring output behavior).
- As $\lambda \to \infty$, we prioritize smooth output transitions at any path length cost.

This is conceptually analogous to the principle of least action in physics. We seek "efficient" paths through activation space that balance distance traveled against output stability.

2.3 Choice of Norm

We use the L_2 (Euclidean) norm for several reasons reflecting epistemic humility:

1. Strict convexity: ensures points along the path are evenly spaced.
2. Isotropy: assumes no privileged directions in activation space. We know features are privileged directions, but we don't know a priori which directions those are.
3. Flatness: assumes no characteristic length scale (infinite radius of curvature). Defining curvature would require choosing a length scale we have no basis for.

3 Results

3.1 Core Finding: Smooth Paths Exist

For the prompt pair "The house was big" → "The house was in" (which shows a sharp boundary along slerp), path optimization finds a much smoother path. Figure 2 shows the relative distance profile: slerp exhibits the characteristic S-curve (flat plateaus with sharp transition), while the optimized path is nearly linear.

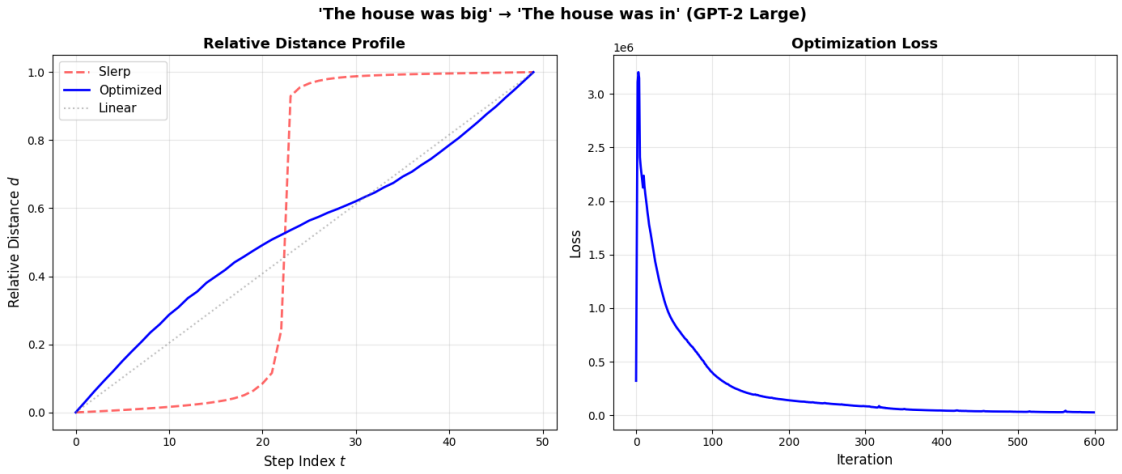


Figure 2: Left: Relative distance $d = \|x - A\|/(\|x - A\| + \|x - B\|)$ along slerp (red) and the optimized path (blue). The slerp path shows the characteristic plateau-boundary-plateau pattern. The optimized path is nearly linear. Right: Optimization loss over iterations.

3.2 The Cost of Smoothness

The smooth path is approximately 20 times longer in activation space than slerp. Figure 3 shows the step lengths and cumulative distance along both paths. The optimized path takes large steps in the transition region (steps 15-35), winding through activation space to avoid the high-sensitivity region that slerp crosses directly.

3.3 Jacobian Analysis

To understand why smooth paths require such long detours, we analyze the directional sensitivity of the model at different points in activation space. The directional sensitivity in direction v is $\|J \cdot v\|$, where J is the Jacobian of the model output with respect to activations.

Figure 4 shows the distribution of sensitivities across 100 random directions at the starting point ("The house was big"). The slerp direction has sensitivity 1.6 (86th percentile), while the optimized path tangent has sensitivity 3.4 (highest). Most random directions have lower sensitivity (median 1.2).

Figure 5 shows how sensitivity evolves along both paths. Along the slerp path (top left), sensitivity in the slerp tangent direction spikes dramatically at the boundary (step ∼23), reaching values above 6000. Along the optimized path (top right), sensitivity in the optimized tangent direction stays moderate throughout, remaining below 25 even in the transition region.

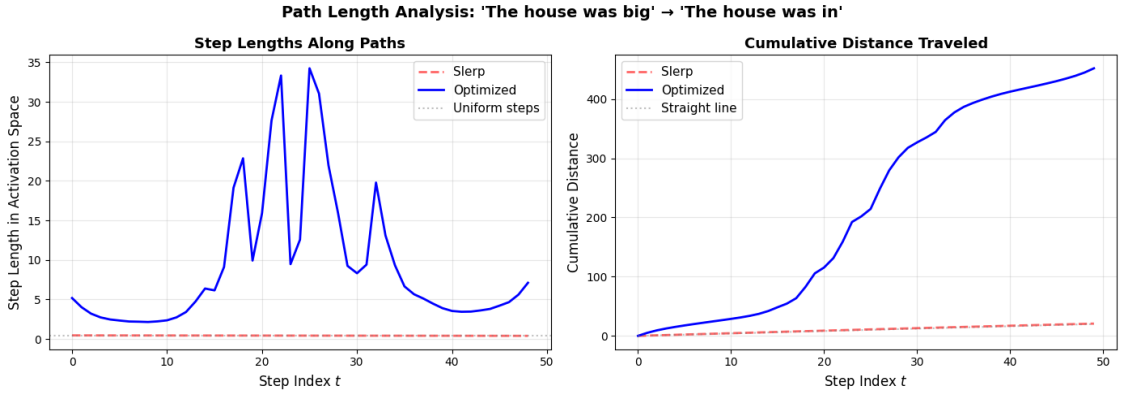


Figure 3: Left: Step length in activation space at each point along the path. Slerp has uniform step sizes; the optimized path has highly variable steps, with large excursions in the transition region. Right: Cumulative distance traveled. The optimized path is roughly 20 times longer.

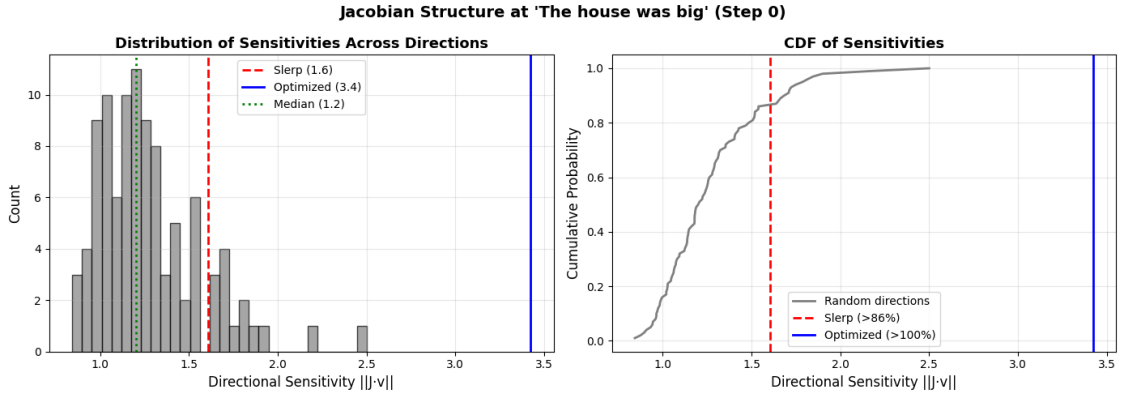


Figure 4: Distribution of directional sensitivities at step 0 ("The house was big"). Left: histogram. Right: CDF. The slerp direction (red) is at the 86th percentile; the optimized path direction (blue) has the highest sensitivity of all directions tested.

The middle row shows output change per step: slerp has a massive spike at the boundary, while the optimized path maintains roughly constant output change. The bottom row compares the product of sensitivity and distance to actual output velocity, showing that the first-order approximation (output change $\approx$ sensitivity $\times$ distance) is reasonable but not exact at the step sizes used.

3.4 Jacobian Eigenspectrum

The directional sensitivity analysis shows that some directions are more sensitive than others, but does not characterize the full structure. To understand how many directions have high sensitivity at a given point, we analyze the eigenspectrum of $J^T J$, where J is the Jacobian of output logits with respect to activations.

The eigenvectors of $J^T J$ are the activation-space directions ordered by sensitivity, and the eigenvalues are the squared singular values of J. For a perturbation v, the output change is $\|Jv\|$; maximizing $\|Jv\|^2 = v^T J^T J v$ subject to $\|v\| = 1$ gives the eigenvalue problem for $J^T J$.

We computed the top 100 eigenvalues at three points along the interpolation: start ("The house

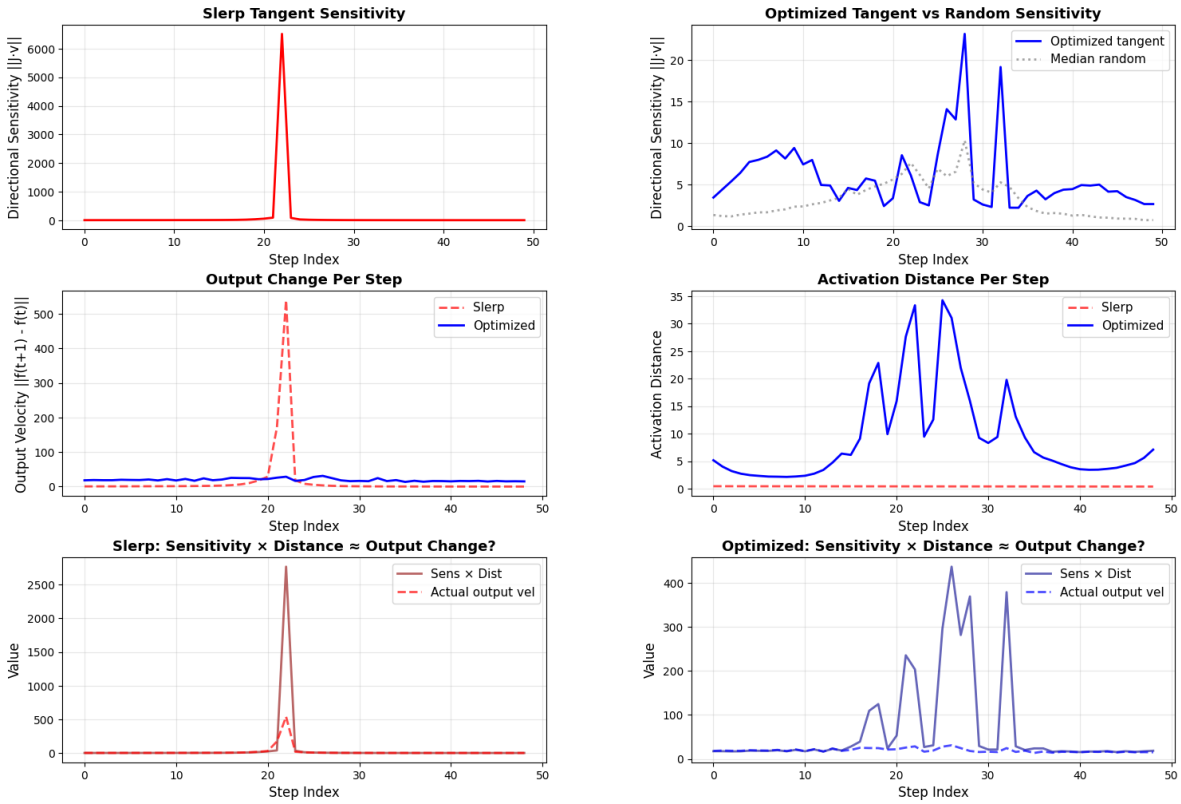


Figure 5: Complete analysis of both paths. Top row: directional sensitivity along each path. Middle row: output change per step and activation distance per step. Bottom row: comparison of sensitivity × distance to actual output velocity.

was big"), boundary (point of maximum directional sensitivity), and end ("The house was in"). We used Lanczos iteration, an algorithm that computes extremal eigenvalues via repeated matrix-vector products without materializing the full matrix. Each iteration requires one Jacobian-vector product, which autodiff provides efficiently.

Figure 6 shows the results:

1. At all three points tested, the spectrum is sparse: one singular value is ∼10× larger than the second. Sensitivity is dominated by a single direction.

2. The top singular value at the boundary (∼4300) is ∼10× larger than at the plateaus (∼400–600).

3. The top singular vectors at different points have cosine similarities of 0.11–0.25, which is 5–10× higher than expected for random vectors in 1280 dimensions (∼0.02), but far from aligned.

4. The interpolation direction aligns with the top singular vector at the boundary (cosine similarity 0.66) but not at the start (0.06).

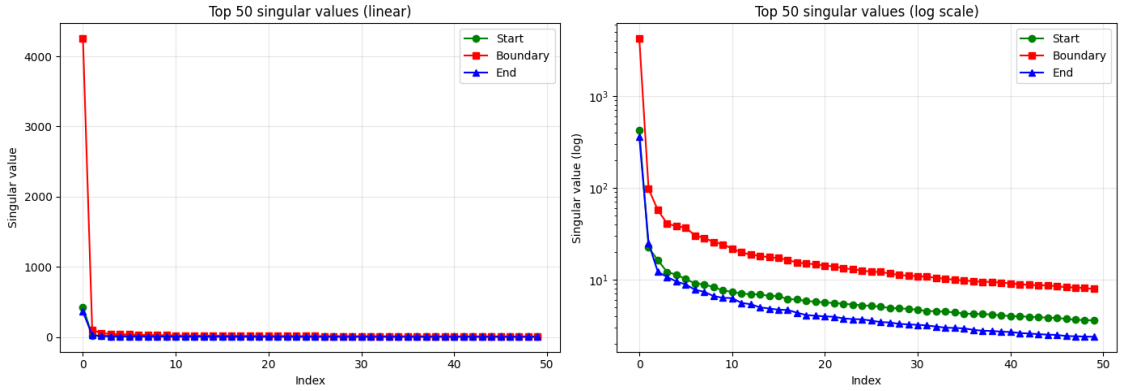


Figure 6: Top 50 singular values at start, boundary, and end points. Left: linear scale. Right: log scale.

4 Open Questions

1. The sparse eigenspectrum (one dominant direction) was observed at all three points tested. Does this hold more generally across activation space?

2. What determines the orientation of the top singular vector at a given point? Is it interpretable?

3. How does this connect to steering? If steering works by pushing activations from one region to another, and the model has high sensitivity along direct paths between semantically different prompts, this might explain why steering magnitude matters so much.